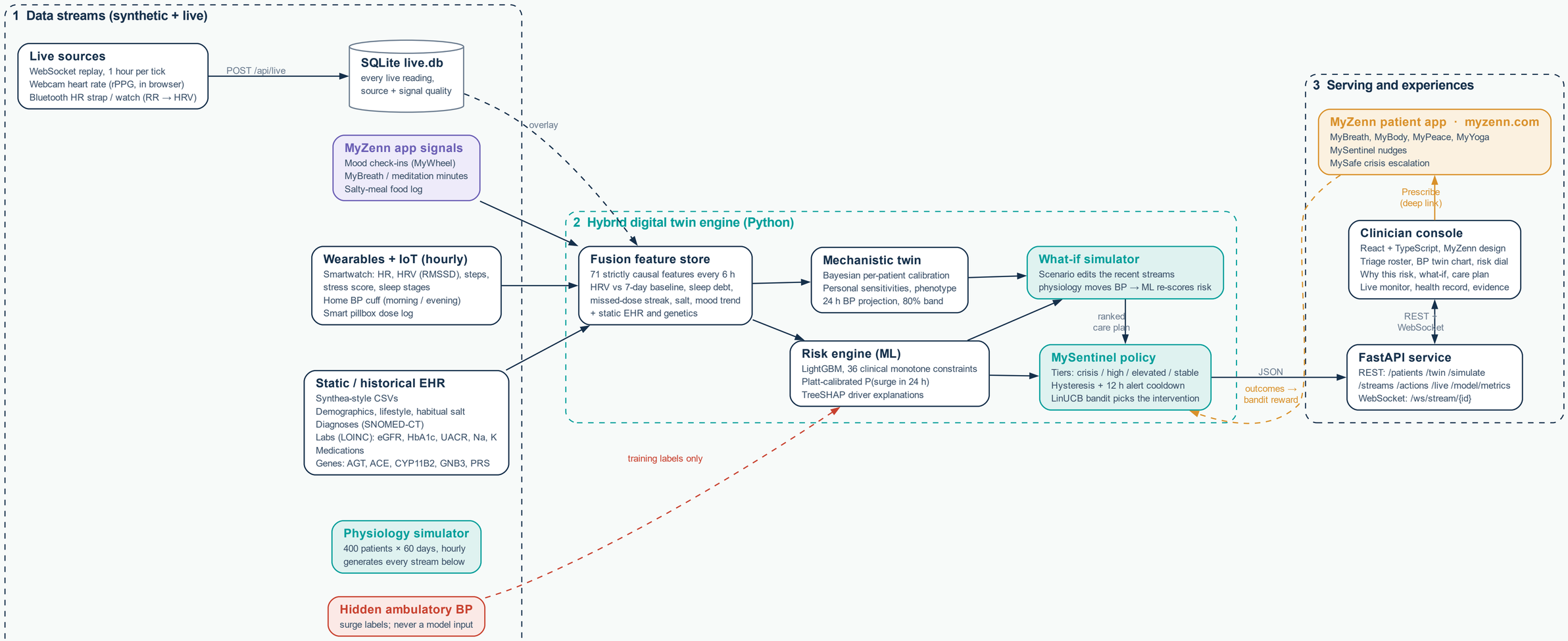
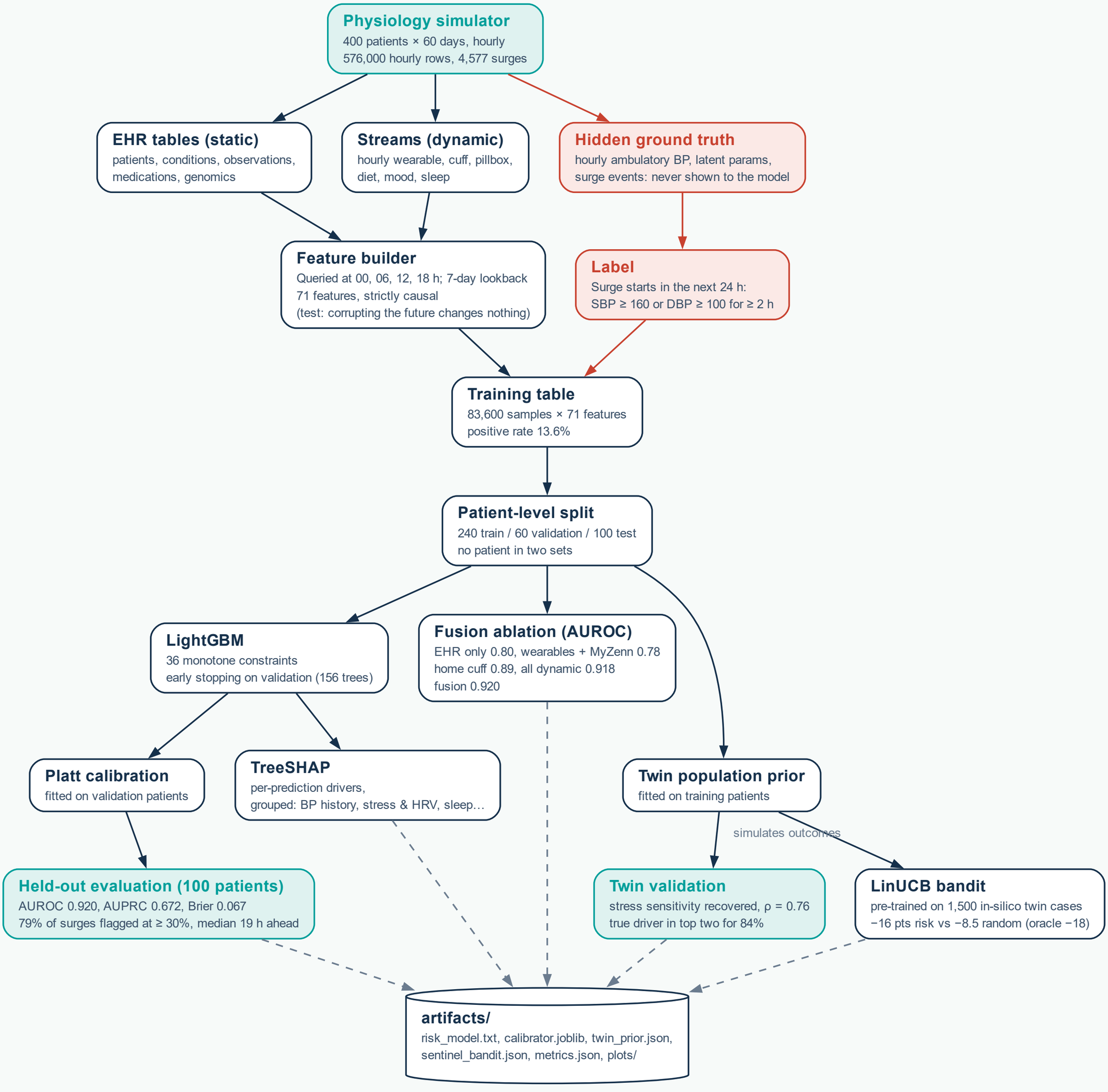
HridayTwin: system architecture

Two data streams fused into a hybrid digital twin; a doctor-facing console; interventions delivered through MyZenn.



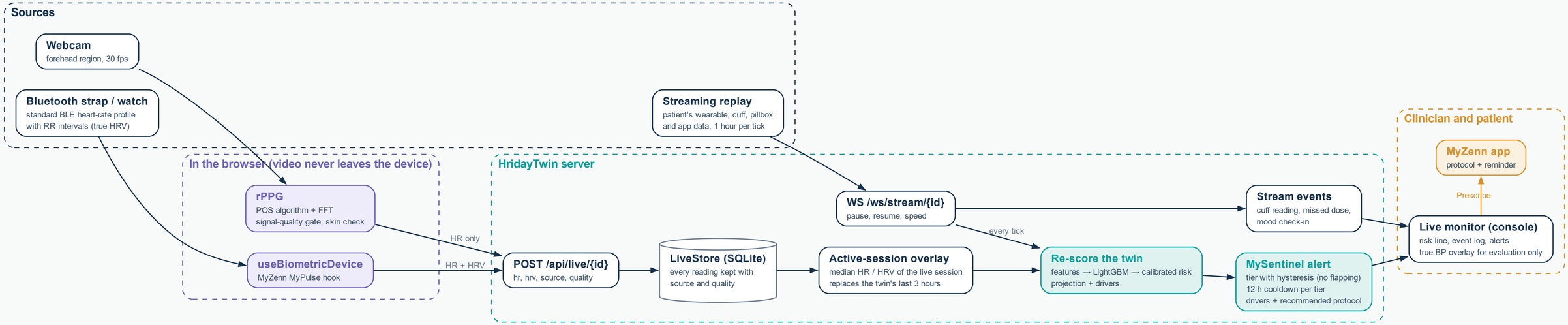
Data fusion and model training pipeline

How the synthetic cohort becomes a calibrated 24-hour surge predictor, with every number from held-out patients.



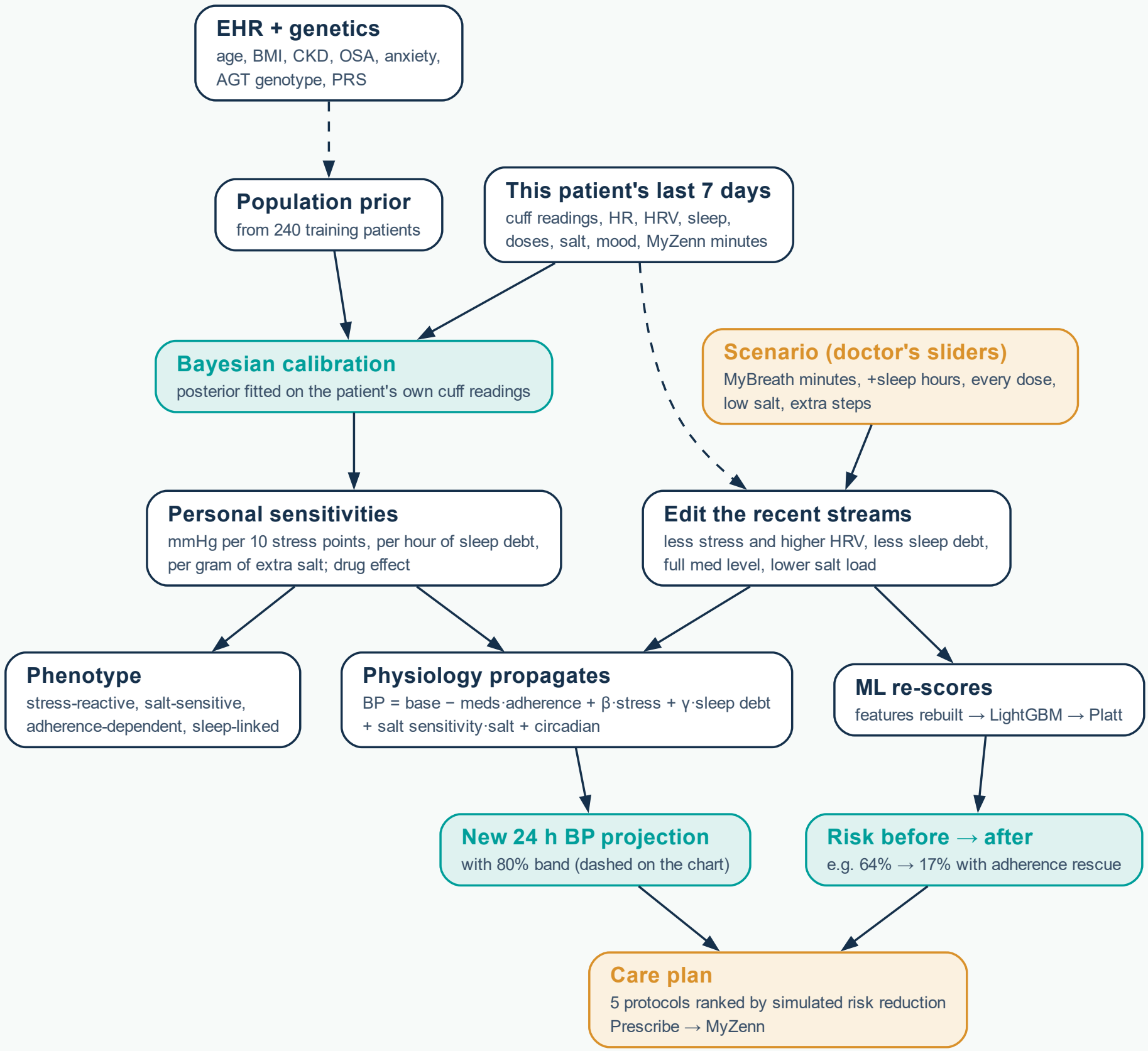
Real-time data flow

Three live sources feed one twin. Every reading is stored, and the risk updates within seconds.



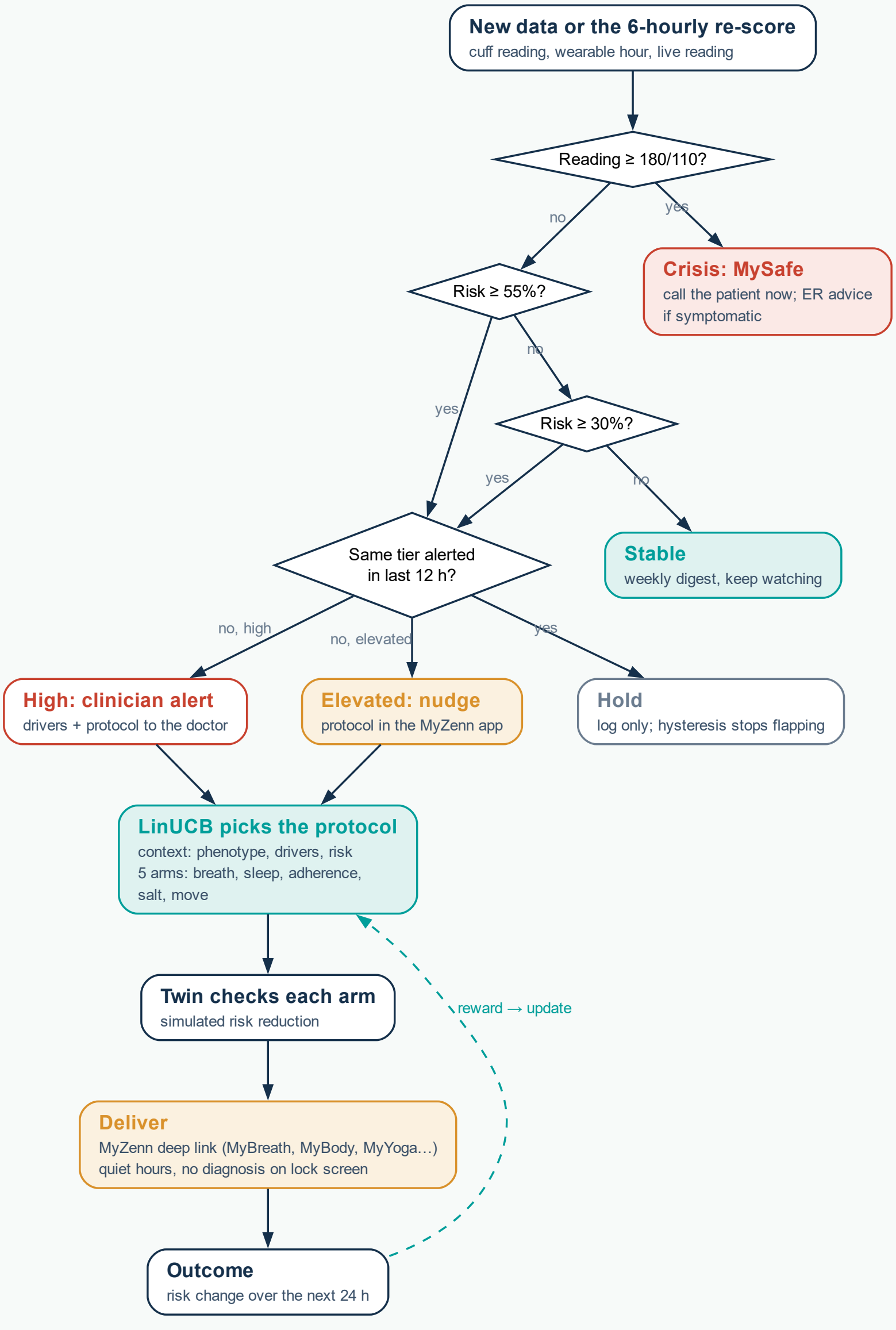
The hybrid twin: ML for accuracy, physiology for what-if

A scenario changes the patient's recent behaviour; the physiology moves blood pressure; the ML model re-scores the risk.



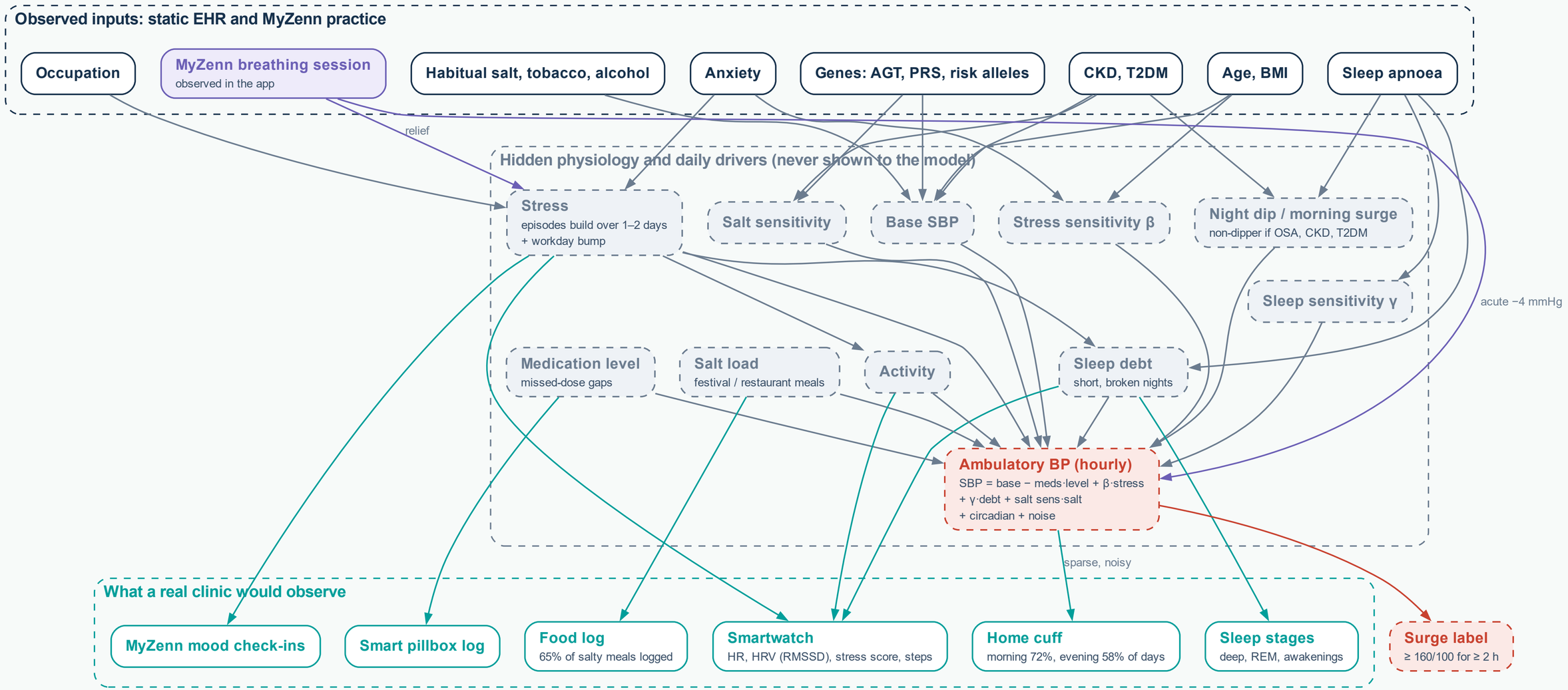
MySentinel: from a risk score to an action

Safety rules first, then the learned policy. Thresholds from twin_engine/config.py.



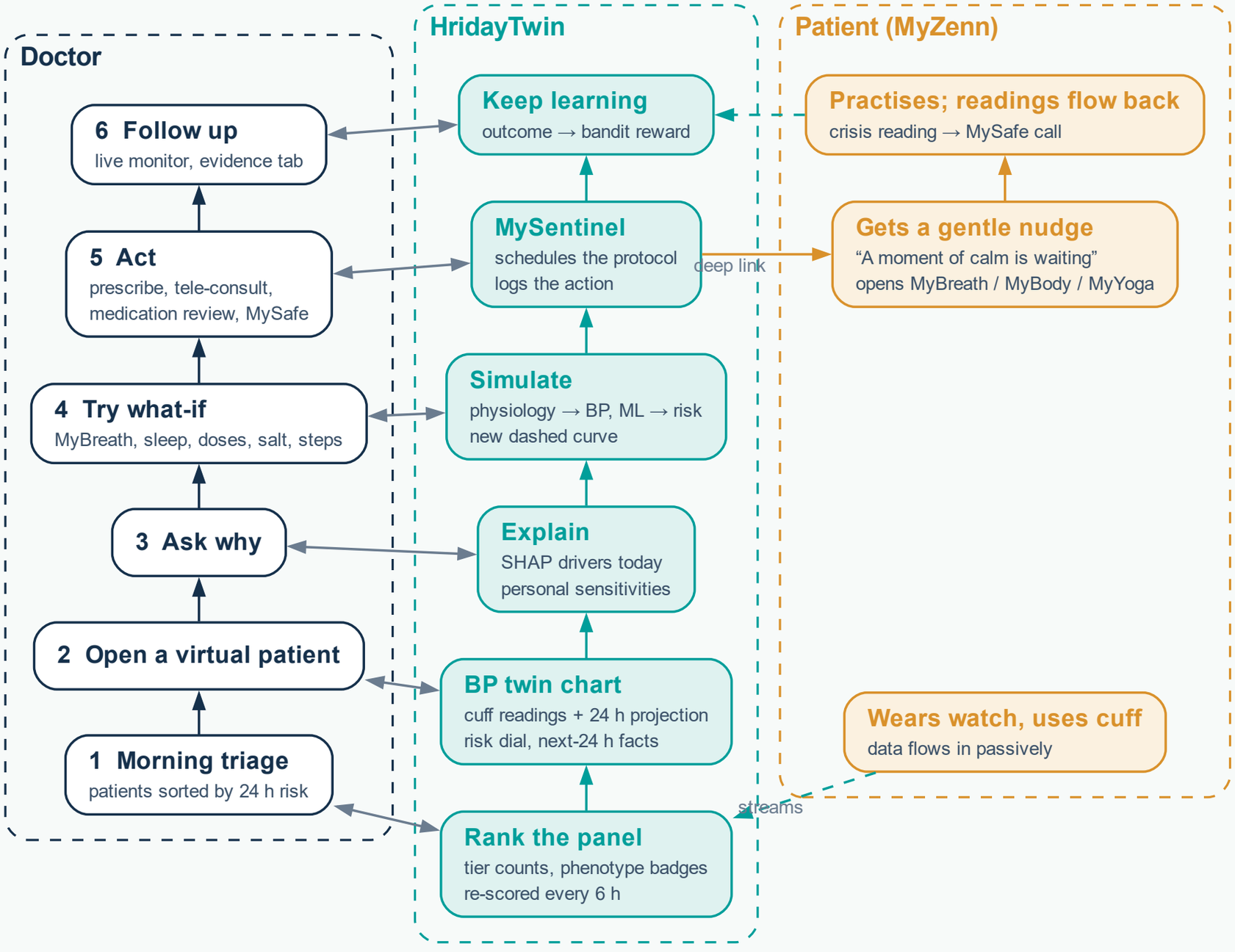
The causal model behind the synthetic cohort

Why every surge has a known cause. Dashed = hidden from the model; solid = what a real clinic would observe.



How a doctor works with the virtual patient

One morning in the HridayTwin console, and what the patient sees in MyZenn.



Privacy and deployment

Synthetic data only for training. On live devices, only derived numbers leave the phone or laptop.

